MIL-R-12934/30D
14 March 1974
SUPERSEDING
MIL-R-12934/30C
30 March 1972

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND, PRECISION,

STYLE RR3900

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RR3900, variable, wire-wound, precision resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

Single-section (cup) resistors

RR3900	A	2	A	7	J	101
Style (1.2.1.1)	Function and shaft length (1.2.1.2)	Class (1.2.1.3)	Resistance temperature characteristic (1.2.1.4)	Rotational life characteristic (1.2.1.5)	Function conformity tolerance and resistance tolerance characteristic (1.2.1.6)	Resistance (1.2.1.7)

1.2.1.1 Style RR3900. Style RR3900 is applicable to this specification.

1.2.1.2 Function and shaft length. The function and shaft length applicable to this specification shall be identified by a single symbol in accordance with table I.

TABLE I. Function and shaft length.

Symbol	Shaft length Tolerance ±1/32 (.79)	Function
A - - - - -	3/8 (9.53)	Linear
B - - - - -	1/2 (12.70)	Linear
C - - - - -	5/8 (15.88)	Linear
D - - - - -	3/4 (19.05)	Linear
E - - - - -	7/8 (22.23)	Linear
F - - - - -	1 (25.40)	Linear

1.2.1.3 Class. The class applicable to this specification shall be symbols 2 and 3.

1.2.1.4 Resistance temperature characteristic. The resistance temperature characteristics applicable to this specification are symbols A to H, inclusive.

1.2.1.5 Rotational life characteristics. The rotational life characteristics applicable to this specification are symbols 7, 8, and 9. This potentiometer is capable of operation up to 100 RPM.

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1.2.1.6 Function conformity tolerance and resistance tolerance characteristics. The function conformity tolerance and resistance tolerance characteristics applicable to this specification are symbols G to L, inclusive and S to X, inclusive.

1.2.1.7 Resistance. The nominal total resistance values applicable to this specification shall be in accordance with table II.

TABLE II. Nominal total resistance and nominal resolution characteristics.

Nominal total resistance value	Nominal resolution
(Ohms)	(Percent)
100	.075
200	.061
500	.048
1,000	.038
2,000	.035
5,000	.028
10,000	.031
20,000	.024
50,000	.018
100,000	.014

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-R-12934 - Resistors, Variable, Wire-wound, Precision.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-12934 and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Minimum resistance. Minimum total resistance value shall be 100 ohms.

3.4 Maximum resistance. Maximum total resistance value shall be 100,000 ohms.

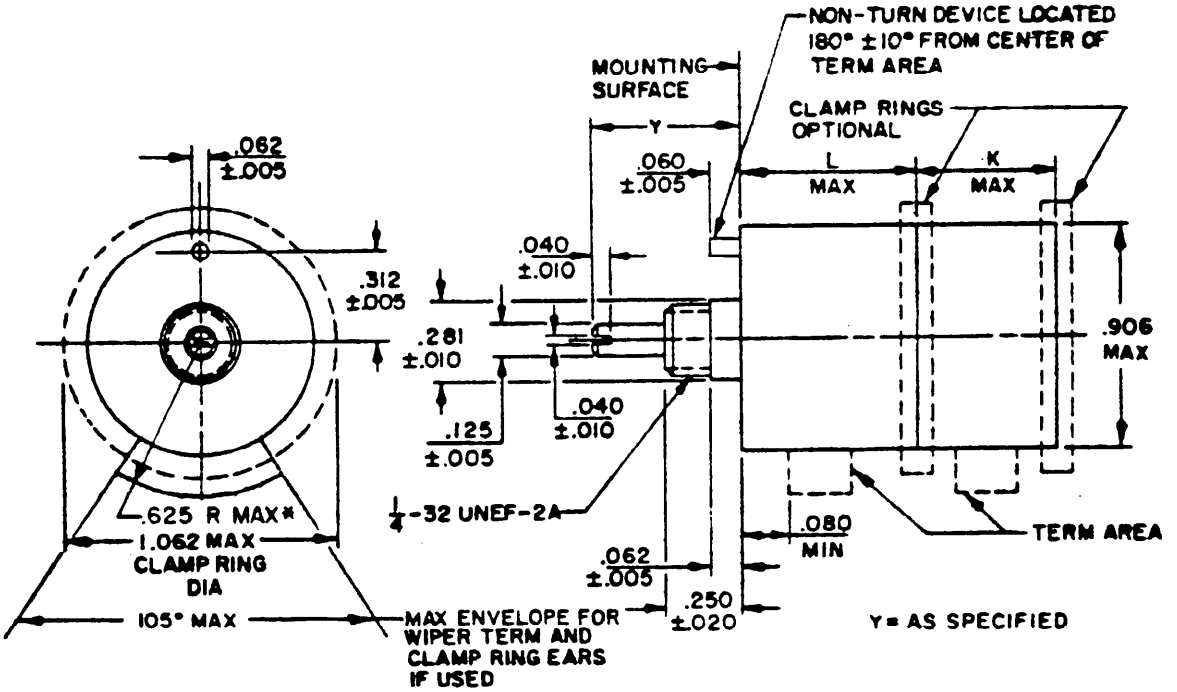
3.5 Mechanical travel. Mechanical travel shall be $1,800\ ^{+10^{\circ}}_{-0^{\circ}}$.

3.6 Actual electrical travel. The actual electrical travel shall be $1,800\ ^{+4^{\circ}}_{-0^{\circ}}$.

3.7 Ganged cups. There shall be no more than 4 cups ganged.

3.8 Phasing. For independent linearity the phasing between the cups shall be within ±1°. This shall be measured, with respect to the first cup, starting from the counterclockwise end point.

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INCHES	MM	INCHES	MM
.005	.13	.125	3.18
.010	.25	.250	6.35
.020	.51	.281	7.14
.040	1.02	.312	7.92
.060	1.52	.625	15.88
.062	1.57	.906	23.01
.080	2.03	1.062	26.97

Style	Turns	L (max)	K (max)	Maximum continuous working voltage (volts)	Power rating (watts) 1/	Maximum starting and running torque (ounce-inch)				Stop-torque (pound-inch)
						Five-turn single cup		Per each additional five-turn cup		
						starting	running	starting	running	
RR3900	5	1.219 (30.96)	.968 (24.59)	500	1.5	.5	.4	.4	.3	3

1/ When 5 turn units are ganged all cups will be 75% of rated wattage.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Style RR3900.

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3.9 Function. The type of function applicable to this specification is linear function (independent linearity).

3.10 Weight. The maximum weight shall be .06 pound (27.22 grams), and for each additional cup .04 pound (18.14 grams).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-12934 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-12934.

6. NOTES

6.1 The notes specified in MIL-R-12934 are applicable to this specification.

6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy -
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-12934/30D RESISTORS, VARIABLE, WIRE-WOUND, PRECISION, STYLE RR3900

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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